Tokenized Option Ladder: Mathematical Formulation

1 Setup and the Splitting Primitive

1.1 Mechanism and Prerequisites

Maturity and clearing. A position is opened against collateral and settled at a single maturity m. At maturity an oracle posts the terminal underlying price X_m *once*; every tranche then clears to a closed-form cash (or collateral) amount. The oracle is therefore needed only at this clearing instant, not during the life of the position.

Two collateral states. Every construction starts from one of two collateral states: one unit of the underlying T (terminal value X_m), or cash C (in the put-side base split $C = S_0$).

Numeraire. Payoffs are written in cash / numeraire value by default; collateral-unit forms (underlying-T units, cash units) are given where useful.

Scope. This section is a payoff *decomposition*, not a pricing model. Premiums are set by the market; the math here only guarantees that the tranches partition the collateral's terminal value exactly.

1.2 The Splitting Primitive

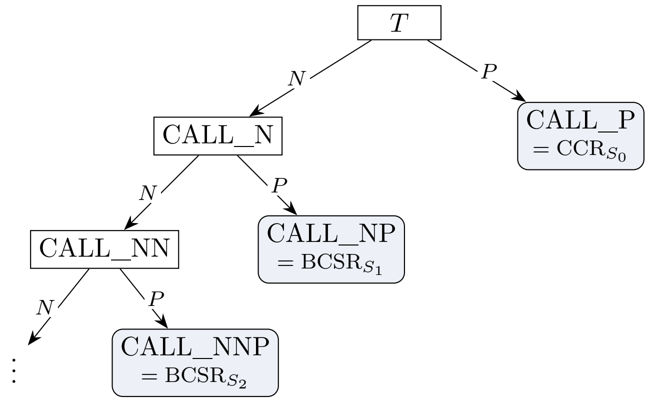
At a chosen strike S, any collateral position splits into an *obligation residual* P and a *right* N,

$$\text{Collateral} = \underbrace{P}_{\text{residual}} + \underbrace{N}_{\text{right}}, \qquad P(X_m) + N(X_m) = (\text{terminal value of the collateral}) \ \ \forall X_m.$$

This conservation identity is what makes every tranche *fully collateralized*: the pieces always sum back to the collateral that backs them.

1.3 Tranche-Address Notation (the P/N Tree)

The suffix on each label is the path taken through the recursion. Starting from the collateral, every split sends the obligation residual to a leaf (P, a tranche the holder keeps) and keeps splitting the right (N):



The strike ladder *is* this recursion. Reading down the right edges gives the menu of tranches the protocol mints—CALL_P (the covered-call residual CCR_{S_0}), then CALL_NP, CALL_NNP, $\ldots$ (the bull-spread residuals $\mathrm{BCSR}_{S_1}, \mathrm{BCSR}_{S_2}, \ldots$)—while the left spine carries the remaining call right down to the tail call CALL_{S_n}. The running index i flattens the repeated-N path, so BCSR_{S_i} is the residual produced at the i-th split.

1.4 Notation

Symbol	Meaning
X	Spot price of the underlying (running).
X_m	Terminal underlying price at maturity m, in cash/numeraire units; the one value the oracle posts at clearing.
T	One unit of underlying collateral; terminal cash value X_m.
C	Cash collateral; in the put-side base split $C = S_0$.
S	A strike price (generic).
S_i	Strike of the i-th split, $i = 0, \ldots, n$; S_0 is the base strike. Call ladder: $S_0 < S_1 < \cdots < S_n$; put ladder: $S_0 > S_1 > \cdots > S_n$.
N	Right leg of a split: the long option right that is split off (+CALL or +PUT); the only leg that splits again, i.e. the recursing branch of the tree.
P	Obligation-residual leg of a split: the collateral the writer keeps after the right N is split off; it carries the short-option obligation. Every split obeys collateral $= P + N$.
CALL_{S_i}	Tokenized call position struck at S_i (a token, not a traded option); terminal value $\max(0, X_m - S_i)$.
PUT_{S_i}	Tokenized put position struck at S_i (a token, not a traded option); terminal value $\max(0, S_i - X_m)$.
$\pm\mathrm{CALL}_{S_i}$, $\pm\mathrm{PUT}_{S_i}$	Long (+) / short (−) the corresponding token.
CC	Covered call (strategy): hold the underlying and write a call against it, capping the upside above the strike in return for the premium.
CCR_{S_0} (CALL_P)	Covered-call residual token: the base-split residual leg $T - \mathrm{CALL}_{S_0}$. Exists only at the base strike S_0.

CSP	Cash-secured put (strategy): hold cash and write a put fully backed by that cash, taking on the obligation to buy the underlying below the strike.
CSPR_{S_0} (PUT_P)	Cash-secured-put residual token: the base-split residual leg $C - \mathrm{PUT}_{S_0}$. Exists only at S_0.
BCS	Bull call spread (strategy): long $\mathrm{CALL}_{S_{i-1}}$, short CALL_{S_i} (the i-th rung), a bounded bullish payoff between the two strikes.
BCSR_{S_i} (CALL_NP)	Bull-call-spread residual token at the i-th split ($i = 1, \ldots, n$): the residual leg of the bull call spread BCS, between strikes S_{i-1} and S_i.
BPS	Bear put spread (strategy): long $\mathrm{PUT}_{S_{i-1}}$, short PUT_{S_i} (the i-th rung), a bounded bearish payoff between the two strikes.
BPSR_{S_i} (PUT_NP)	Bear-put-spread residual token at the i-th split ($i = 1, \ldots, n$): the residual leg of the bear put spread BPS, between strikes S_{i-1} and S_i.

2 Route Strategy Map

StrikeX route strategy map

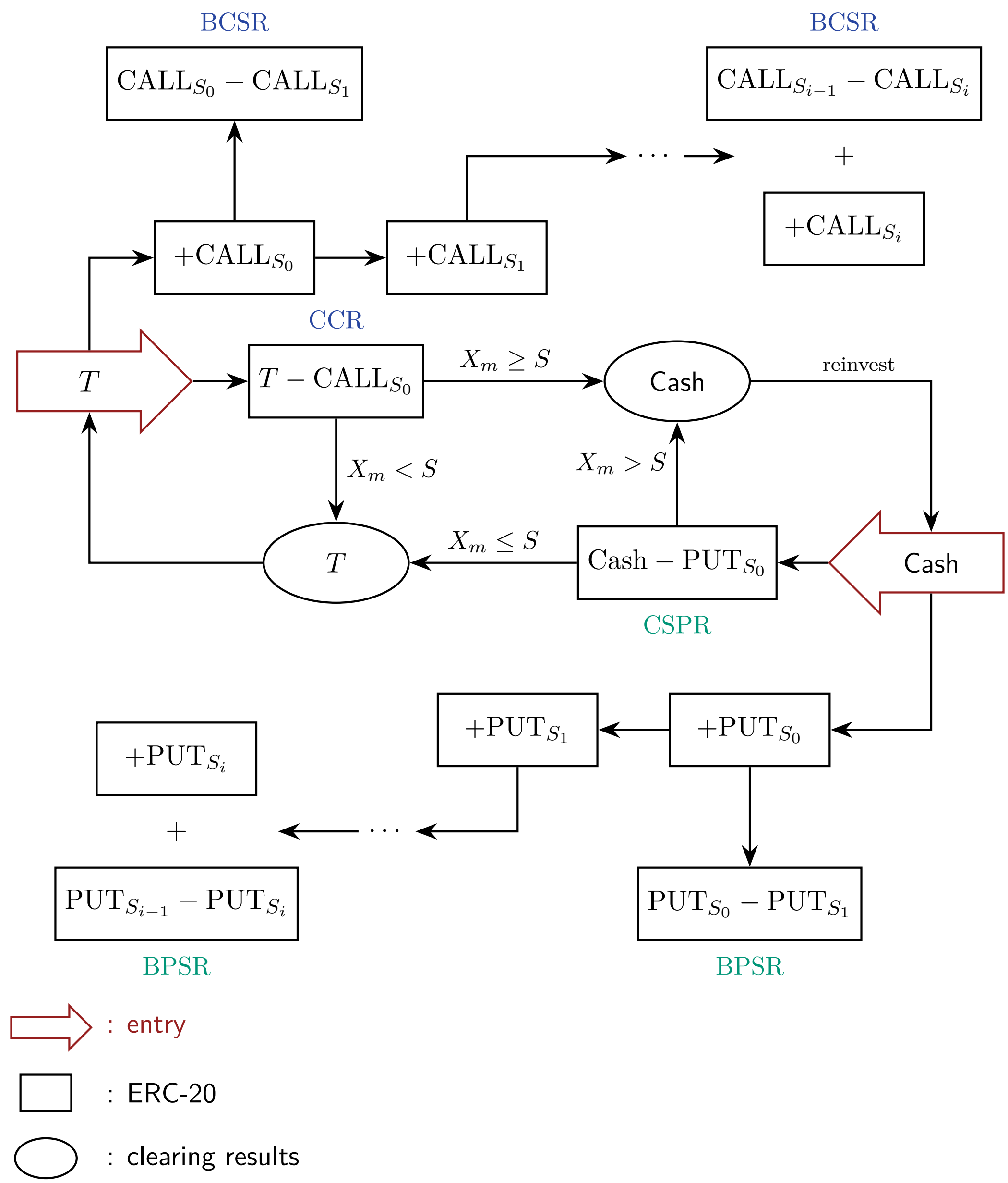


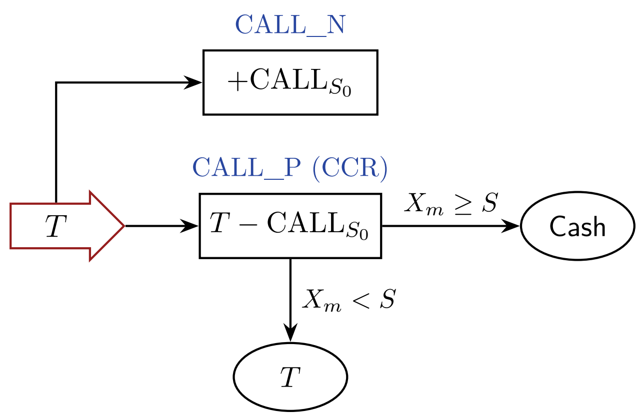
Figure 1: Route strategy map. From each collateral state, the base split yields an obligation residual (P, the $\mathrm{CCR}_{S_0}/\mathrm{CSPR}_{S_0}$ token) and a right (N); the residuals settle to cash or T at maturity according to X_m vs. the strike S, while the rights recurse along the strike ladder into the vertical-spread residuals BCSR/BPSR.

Figure 1 indexes the whole construction: each strategy of Sections 3–4 is one labeled route on this map, and Section 5 reads the same graph as a reinvestment cycle.

3 Long Underlying Assets

A long position in the underlying T (terminal value X_m) is tokenized into a ladder of call-side tranches. The base split (Section 3.1) carves T into a long-call token and a covered-call residual; the recursive split (Section 3.2) carves the long-call token into bull-call-spread tranches along the strike ladder; and the whole ladder conserves back to X_m (Section 3.3).

3.1 Call-Side Base Split & Covered Call



We start *long the underlying*: one unit of collateral T, worth X_m at maturity. At the base strike S_0 this long position is tokenized into two ERC-20 tranches—a right N and a residual P:

$$T = (N, P) \;\longmapsto\; \text{CALL\_N (long-call token)} \;+\; \text{CALL\_P (covered-call residual)}.$$

(a) Splitting. At S_0 the underlying splits into the covered-call residual (the underlying minus the written call) and the call right:

$$T = (T - \mathrm{CALL}_{S_0}) + \mathrm{CALL}_{S_0}, \tag{1}$$

$$T = \mathrm{CALL\_P}_{S_0} + \mathrm{CALL\_N}_{S_0}. \tag{2}$$

$\mathrm{CALL\_P}_{S_0} = T - \mathrm{CALL}_{S_0}$ is the covered-call leg—long the underlying, short the S_0 call, so its upside is capped at S_0. $\mathrm{CALL\_N}_{S_0} = \mathrm{CALL}_{S_0}$ is the long-call leg—the uncapped upside above S_0, collateralized by the residual holder.

(b) Settlement. At maturity each token clears to a closed form:

$$\mathrm{CALL\_N}_{S_0}(X_m) = (X_m - S_0)^+ = \max(0, X_m - S_0), \tag{3}$$

$$\mathrm{CALL\_P}_{S_0}(X_m) = X_m - (X_m - S_0)^+ = \min(X_m, S_0). \tag{4}$$

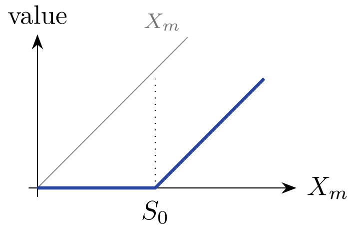
The covered-call leg converts the underlying into cash S_0 when the call is assigned ($X_m \geq S_0$) and keeps the underlying T, worth X_m, otherwise. The two legs conserve:

$$\mathrm{CALL\_P}_{S_0}(X_m) + \mathrm{CALL\_N}_{S_0}(X_m) = \min(X_m, S_0) + (X_m - S_0)^+ = X_m. \tag{5}$$

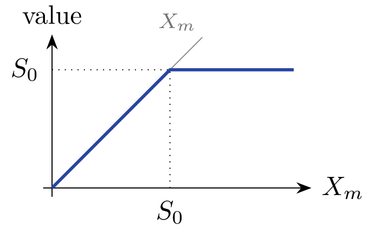
In underlying-collateral units ($X_m > 0$),

$$\mathrm{CALL\_P}^{T}_{S_0} = \min\Big(1, \tfrac{S_0}{X_m}\Big)\, T, \qquad \mathrm{CALL\_N}^{T}_{S_0} = \max\Big(0, 1 - \tfrac{S_0}{X_m}\Big)\, T.$$

(c) Settlement charts. The two legs are the standard long-call and covered-call payoffs (USD settlement value vs. underlying price X_m):

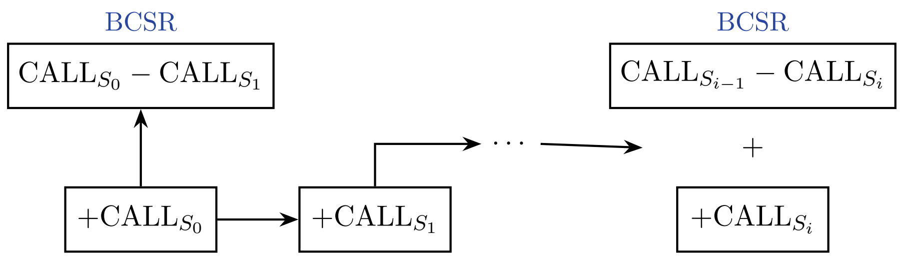


long-call leg CALL_N



covered-call leg CALL_P

3.2 Call-Side Recursive Split & Bull Call Spread



The long-call token CALL_N is itself split at the next rung. For a call ladder with increasing strikes $S_0 < S_1 < \cdots < S_n$, each lower-strike call right splits into a capped bull-call-spread residual plus the higher-strike call carried further down the ladder. The call right is tokenized into a further right N and a residual P:

$$\mathrm{CALL\_N} = (N, P) \;\longmapsto\; \mathrm{CALL\_NN}\ (\text{higher-strike tail call}) \;+\; \mathrm{CALL\_NP}\ (\text{bull-call-spread residual}).$$

(a) Splitting. For $i = 0, \ldots, n-1$, the call right $\mathrm{CALL\_N}_{S_i}$ is split at the next strike S_{i+1}:

$$\mathrm{CALL\_N}_{S_i} = (\mathrm{CALL}_{S_i} - \mathrm{CALL}_{S_{i+1}}) + \mathrm{CALL}_{S_{i+1}}, \tag{6}$$
$$\mathrm{CALL\_N}_{S_i} = \mathrm{CALL\_NP}_{S_{i+1}} + \mathrm{CALL\_NN}_{S_{i+1}}. \tag{7}$$

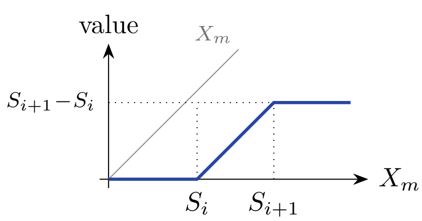
$\mathrm{CALL\_NP}_{S_{i+1}}$ is the bull-call-spread residual (long the S_i call, short the S_{i+1} call); $\mathrm{CALL\_NN}_{S_{i+1}}$ is the higher-strike tail call — itself the call right $\mathrm{CALL\_N}_{S_{i+1}}$ handed to the next rung, which splits again.

(b) Settlement.

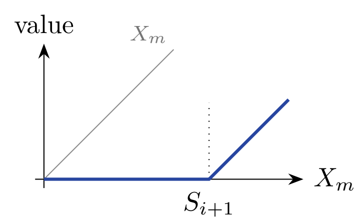
$$\mathrm{CALL\_NP}_{S_{i+1}}(X_m) = (X_m - S_i)^+ - (X_m - S_{i+1})^+ = \min(\max(0, X_m - S_i),\, S_{i+1} - S_i)\,, \tag{8}$$
$$\mathrm{CALL\_NN}_{S_{i+1}}(X_m) = (X_m - S_{i+1})^+ = \max(0, X_m - S_{i+1}). \tag{9}$$

Since $S_{i+1} - S_i > 0$, the spread residual is non-negative and capped at $S_{i+1} - S_i$.

(c) Settlement charts. The bull-spread leg is capped between adjacent strikes; the tail call is an ordinary long call struck at S_{i+1}:



bull-spread leg CALL_NP



tail-call leg CALL_NN

3.3 Call-Side Conservation: Combined Chart and Table

After recursively splitting to S_n, the full call-side ladder identity is

$$X_m = \mathrm{CCR}_{S_0}(X_m) + \sum_{i=1}^{n} \mathrm{BCSR}_{S_i}(X_m) + \mathrm{CALL}_{S_n}(X_m), \tag{10}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \left[(X_m - S_{i-1})^+ - (X_m - S_i)^+\right] + (X_m - S_n)^+ , \tag{11}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \min(\max(0, X_m - S_{i-1}), S_i - S_{i-1}) + \max(0, X_m - S_n). \tag{12}$$

The right-leg sum is telescoping:

$$\sum_{i=1}^{n} \left[(X_m - S_{i-1})^+ - (X_m - S_i)^+\right] + (X_m - S_n)^+ = (X_m - S_0)^+ , \tag{13}$$

so $\min(X_m, S_0) + (X_m - S_0)^+ = X_m$: the tranches always sum back to the underlying.

Numerical check. Parameters $S_0 = 2000$, $S_1 = 2500$, with

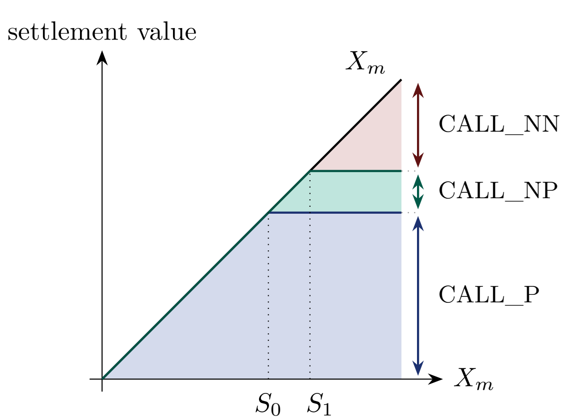
$$\mathrm{CALL\_P} = \min(X_m, 2000), \tag{14}$$

$$\mathrm{CALL\_NP} = \min(\max(0, X_m - 2000), 500) , \tag{15}$$

$$\mathrm{CALL\_NN} = \max(0, X_m - 2500). \tag{16}$$

X_m	CALL_P	CALL_NP	CALL_NN	Total
3600	2000	500	1100	3600
2600	2000	500	100	2600
2500	2000	500	0	2500
2000	2000	0	0	2000
1700	1700	0	0	1700
1500	1500	0	0	1500
1000	1000	0	0	1000

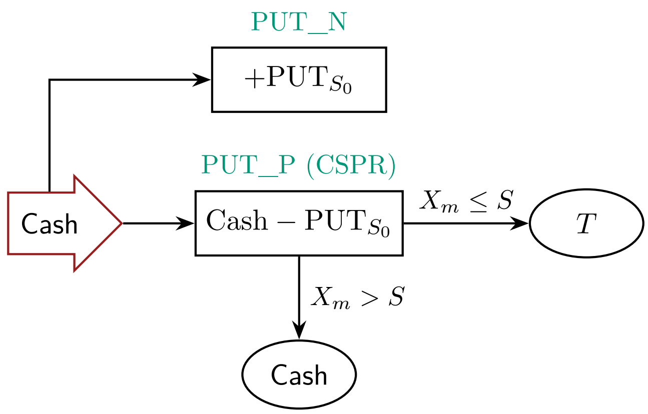
Combined chart. Stacking the tokens at every price fills exactly up to the 45° line X_m — the visual statement of conservation ($S_0 = 2000$, $S_1 = 2500$):



4 Short Underlying Assets

A cash position $C = S_0$ — the bearish, "short" side — is tokenized into a ladder of put-side tranches. The base split (Section 4.1) carves the cash into a long-put token and a cash-secured-put residual; the recursive split (Section 4.2) carves the long-put token into bear-put-spread tranches down the strike ladder; and the whole ladder conserves back to the fixed cash value S_0 (Section 4.3) — not to X_m, the key asymmetry with the call side.

4.1 Put-Side Base Split & Cash-Secured Put



We start with *cash* collateral: $C = S_0$. At the base strike S_0 this cash is tokenized into two ERC-20 tranches—a right N and a residual P:

$$C = (N, P) \;\longmapsto\; \text{PUT\_N (long-put token)} \;+\; \text{PUT\_P (cash-secured-put residual)}.$$

(a) Splitting. At S_0 the cash splits into the cash-secured-put residual (the cash minus the written put) and the put right:

$$C = (C - \mathrm{PUT}_{S_0}) + \mathrm{PUT}_{S_0}, \tag{17}$$
$$C = \mathrm{PUT\_P}_{S_0} + \mathrm{PUT\_N}_{S_0}. \tag{18}$$

$\mathrm{PUT\_P}_{S_0} = C - \mathrm{PUT}_{S_0}$ is the cash-secured-put leg—hold cash, short the S_0 put, so it is obligated to buy the underlying below S_0. $\mathrm{PUT\_N}_{S_0} = \mathrm{PUT}_{S_0}$ is the long-put leg—the bearish downside below S_0, collateralized by the residual holder.

(b) Settlement. At maturity each token clears to a closed form:

$$\mathrm{PUT\_N}_{S_0}(X_m) = (S_0 - X_m)^+ = \max(0, S_0 - X_m), \tag{19}$$
$$\mathrm{PUT\_P}_{S_0}(X_m) = S_0 - (S_0 - X_m)^+ = \min(X_m, S_0). \tag{20}$$

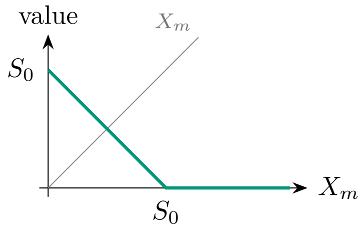
The cash-secured-put leg converts the cash into the underlying T when the put is assigned ($X_m \le S_0$) and keeps the cash S_0 otherwise. The two legs conserve to the fixed cash:

$$\mathrm{PUT\_P}_{S_0}(X_m) + \mathrm{PUT\_N}_{S_0}(X_m) = \min(X_m, S_0) + (S_0 - X_m)^+ = S_0 = C. \tag{21}$$

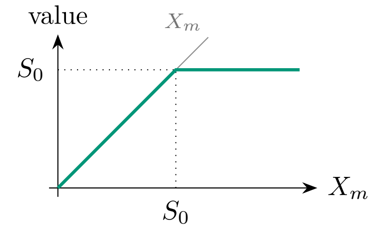
In cash-collateral units ($C = S_0$),

$$\mathrm{PUT\_P}^{C}_{S_0} = \min\Big(1, \tfrac{X_m}{S_0}\Big)\, C, \qquad \mathrm{PUT\_N}^{C}_{S_0} = \max\Big(0, 1 - \tfrac{X_m}{S_0}\Big)\, C.$$

(c) Settlement charts. The two legs are the standard long-put and cash-secured-put payoffs (USD settlement value vs. underlying price X_m):

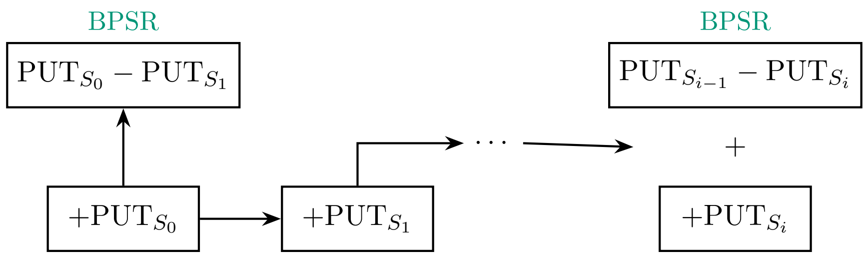


long-put leg PUT_N



cash-secured-put leg PUT_P

4.2 Put-Side Recursive Split & Bear Put Spread



The long-put token PUT_N is itself split at the next rung. For a put ladder with *decreasing* strikes $S_0 > S_1 > \cdots > S_n$, each higher-strike put right splits into a capped bear-put-spread residual plus the lower-strike put carried further down the ladder. The put right is tokenized into a further right N and a residual P:

$$\mathrm{PUT\_N} = (N, P) \;\longmapsto\; \mathrm{PUT\_NN}\ \text{(lower-strike tail put)} \;+\; \mathrm{PUT\_NP}\ \text{(bear-put-spread residual)}.$$

(a) Splitting. For $i = 0, \ldots, n-1$, the put right $\mathrm{PUT\_N}_{S_i}$ is split at the next (lower) strike S_{i+1}:

$$\mathrm{PUT\_N}_{S_i} = (\mathrm{PUT}_{S_i} - \mathrm{PUT}_{S_{i+1}}) + \mathrm{PUT}_{S_{i+1}}, \tag{22}$$

$$\mathrm{PUT\_N}_{S_i} = \mathrm{PUT\_NP}_{S_{i+1}} + \mathrm{PUT\_NN}_{S_{i+1}}. \tag{23}$$

$\mathrm{PUT\_NP}_{S_{i+1}}$ is the bear-put-spread residual (long the S_i put, short the S_{i+1} put); $\mathrm{PUT\_NN}_{S_{i+1}}$ is the lower-strike tail put — itself the put right $\mathrm{PUT\_N}_{S_{i+1}}$ handed to the next rung, which splits again.

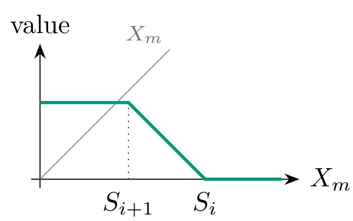
(b) Settlement.

$$\mathrm{PUT\_NP}_{S_{i+1}}(X_m) = (S_i - X_m)^+ - (S_{i+1} - X_m)^+ = \min(\max(0, S_i - X_m),\, S_i - S_{i+1})\,, \tag{24}$$

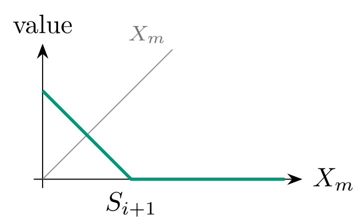
$$\mathrm{PUT\_NN}_{S_{i+1}}(X_m) = (S_{i+1} - X_m)^+ = \max(0, S_{i+1} - X_m). \tag{25}$$

Since the put strikes decrease ($S_i > S_{i+1}$), $S_i - S_{i+1} > 0$: the spread residual is non-negative and capped at $S_i - S_{i+1}$.

(c) Settlement charts. The bear-spread leg is capped between adjacent strikes; the tail put is an ordinary long put struck at S_{i+1}:



bear-spread leg PUT_NP



tail-put leg PUT_NN

4.3 Put-Side Conservation: Combined Chart and Table

After recursively splitting to S_n, the full put-side ladder identity is

$$S_0 = \mathrm{CSPR}_{S_0}(X_m) + \sum_{i=1}^{n} \mathrm{BPSR}_{S_i}(X_m) + \mathrm{PUT}_{S_n}(X_m), \tag{26}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \left[(S_{i-1} - X_m)^+ - (S_i - X_m)^+\right] + (S_n - X_m)^+ , \tag{27}$$

$$= \min(X_m, S_0) + \sum_{i=1}^{n} \min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i) + \max(0, S_n - X_m). \tag{28}$$

The right-leg sum is telescoping:

$$\sum_{i=1}^{n} \left[(S_{i-1} - X_m)^+ - (S_i - X_m)^+\right] + (S_n - X_m)^+ = (S_0 - X_m)^+ , \tag{29}$$

so $\min(X_m, S_0) + (S_0 - X_m)^+ = S_0$: the tranches always sum back to the cash collateral.

Numerical check. Parameters $C = S_0 = 2000$, $S_1 = 1500$, with

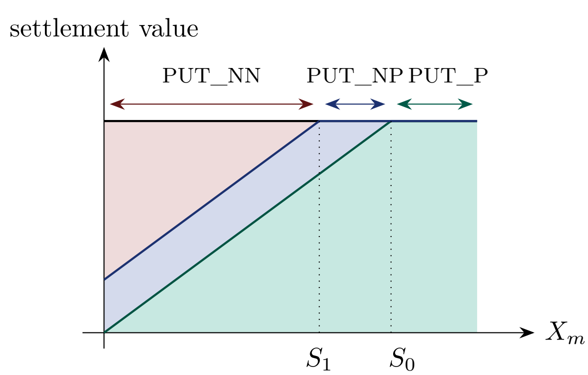
$$\mathrm{PUT\_P} = \min(X_m, 2000), \tag{30}$$

$$\mathrm{PUT\_NP} = \min(\max(0, 2000 - X_m), 500)\,, \tag{31}$$

$$\mathrm{PUT\_NN} = \max(0, 1500 - X_m). \tag{32}$$

X_m	PUT_P	PUT_NP	PUT_NN	Total
3600	2000	0	0	2000
2600	2000	0	0	2000
2500	2000	0	0	2000
2000	2000	0	0	2000
1700	1700	300	0	2000
1500	1500	500	0	2000
1000	1000	500	500	2000
900	900	500	600	2000

Combined chart. Stacking the tokens at every price fills exactly up to the *flat* line S_0 — the put-side total is fixed cash, the key asymmetry with the call side ($S_0 = 2000$, $S_1 = 1500$):



5 Wheeling Strategy

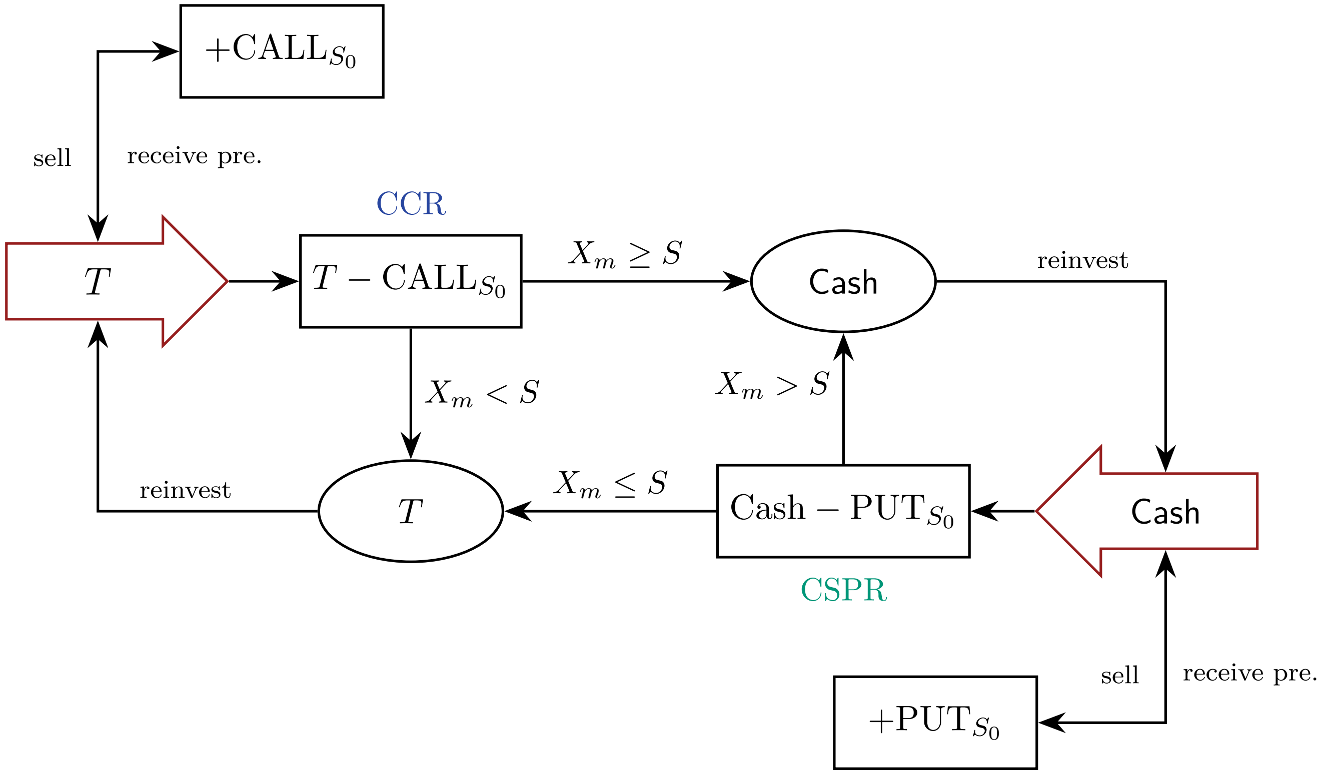


Figure 2: The wheel: a state machine over the underlying state T and the cash state. Red arrows are entry points; each state splits into an obligation residual (P) and a right (N). The covered-call residual (CCR_{S_0}) and its mirror the cash-secured-put residual (CSPR_{S_0}) move collateral between the T and Cash states at maturity according to X_m vs. the strike S; each clearing re-mints collateral into the opposite state. The right legs $+\text{CALL}_{S_0}$ and $+\text{PUT}_{S_0}$ feed the call/put ladders of Sections 3–4.

The wheel cycles collateral between the underlying state T and cash. Each red entry splits a state into a residual (CCR_{S_0} or CSPR_{S_0}) and a right that is sold for premium; at maturity CCR_{S_0} converts T to cash when $X_m \geq S$ (and keeps T otherwise), while its mirror CSPR_{S_0} converts cash to T when $X_m \leq S$ (and keeps cash otherwise). Each clearing re-mints collateral into the opposite state, ready to be re-entered at a new strike—the reinvest loop. The right legs CALL_N/PUT_N feed the ladders of Sections 3–4; closed forms follow below.

Equations referenced in the diagram

Base splits (one maturity step):

$$T = \underbrace{(T - \text{CALL})}_{\text{CALL\_P}} + \underbrace{\text{CALL}}_{\text{CALL\_N}}, \qquad \text{Cash} = \underbrace{(\text{Cash} - \text{PUT})}_{\text{PUT\_P}} + \underbrace{\text{PUT}}_{\text{PUT\_N}}.$$

Entry-split weights (token / cash units; here S denotes the base strike S_0 and X_m the terminal price):

$$\text{CALL\_P} = \min\left(1, \tfrac{S}{X_m}\right) T, \qquad \text{CALL\_N} = \max\left(0, 1 - \tfrac{S}{X_m}\right) T,$$
$$\text{PUT\_P} = \min\left(1, \tfrac{X_m}{S}\right) C, \qquad \text{PUT\_N} = \max\left(0, 1 - \tfrac{X_m}{S}\right) C.$$

Per-bucket spread and tail payoffs:

$$\begin{aligned} \text{CALL\_NP}_{S_i} &= \min\big(\max(0, X_m - S_{i-1}),\ S_i - S_{i-1}\big), & \text{CALL\_NN}_{S_i} &= \max(0, X_m - S_i), \\ \text{PUT\_NP}_{S_i} &= \min\big(\max(0, S_{i-1} - X_m),\ S_{i-1} - S_i\big), & \text{PUT\_NN}_{S_i} &= \max(0, S_i - X_m). \end{aligned}$$

6 Summary

Compact identities

Call side, increasing strikes $S_0 < S_1 < \cdots < S_n$ (conserves to X_m):

$$X_m = \underbrace{\min(X_m, S_0)}_{\text{covered-call residual}} + \sum_{i=1}^{n} \underbrace{\min(\max(0, X_m - S_{i-1}), S_i - S_{i-1})}_{\text{bull call spread residual}} + \underbrace{\max(0, X_m - S_n)}_{\text{tail call}}. \tag{33}$$

Put side, decreasing strikes $S_0 > S_1 > \cdots > S_n$ (conserves to S_0):

$$S_0 = \underbrace{\min(X_m, S_0)}_{\text{cash-secured-put residual}} + \sum_{i=1}^{n} \underbrace{\min(\max(0, S_{i-1} - X_m), S_{i-1} - S_i)}_{\text{bear put spread residual}} + \underbrace{\max(0, S_n - X_m)}_{\text{tail put}}. \tag{34}$$

Tranche reference

Ticker	Meaning	Formula
CALL_P / CCR_{S_0}	Covered-call residual	$\min(X_m, S_0)$
CALL_N	Call right (long call)	$\max(0, X_m - S_0)$
CALL_NP / BCSR_{S_i}	Bull call spread residual	$\min(\max(0, X_m - S_{i-1}),\ S_i - S_{i-1})$
CALL_NN	Higher-strike tail call	$\max(0, X_m - S_i)$
PUT_P / CSPR_{S_0}	Cash-secured-put residual	$\min(X_m, S_0)$
PUT_N	Put right (long put)	$\max(0, S_0 - X_m)$
PUT_NP / BPSR_{S_i}	Bear put spread residual	$\min(\max(0, S_{i-1} - X_m),\ S_{i-1} - S_i)$
PUT_NN	Lower-strike tail put	$\max(0, S_i - X_m)$